

Laboratory Report: EEL2 - No 1

Introduction to Scope Measurements & RC Filters

Institution: Hamburg University of Applied Sciences (HAW Hamburg)

Department: Information Engineering

1. Objectives

This laboratory session focused on the professional operation of a Digital Storage Oscilloscope (DSO) and the experimental analysis of electronic signals. Objectives included:

- Precise measurement of periodic signal parameters (Frequency, Period, Amplitude).
- Verification of RMS (Effective), Mean, and Rectified voltage values across different waveforms.
- Experimental determination of RC filter frequency response and cutoff frequency.
- Observation of phase shift and signal integration in non-sinusoidal waveforms.

2. Equipment Configuration

- **Oscilloscope:** Tektronix TDS 3014B Digital Storage Oscilloscope.
- **Measurement:** Digital Multimeter (Type 18S).
- **Circuit Components:** Resistor $R = 2.2 \text{ k}\Omega$, Capacitor $C = 47 \text{ nF}$.

3. Signal Analysis Results

3.1 Comparison of Waveform Metrics

Signals were generated at 800 Hz with a Peak-to-Peak voltage of 6.0 V. The measured vs. theoretical values are summarized below:

Waveform	Peak-to-Peak (U _{pp})	Measured RMS (U _{eff})	Mean Value
Sine	6.0 V	2.12 V	0.0 V
Square	6.0 V	3.00 V	0.0 V
Triangle	6.0 V	1.73 V	0.0 V

4. RC Filter Performance

4.1 Cutoff Frequency & Phase Response

Based on component values (2.2 kΩ, 47 nF), the theoretical cutoff frequency was identified as ≈ 1.54 kHz. The experiment involved:

- Measuring output gain (U_c/U_e) across a wide frequency spectrum (100 Hz - 100 kHz).
- Using DSO cursor functions to measure time delay (Δt) and calculate phase shift (ϕ).
- Documenting the -20 dB/decade roll-off characteristic typical of first-order low-pass filters.

5. Conclusion

The experimental results successfully validated the theoretical models for signal properties and RC filter behavior. The practical use of DSO triggering and cursor measurements provided high-accuracy data consistent with mathematical predictions for periodic waveforms.

